

# CariSurg MedTech Pathways Research Project

## Preliminary Proposal Template

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### Project Title:

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### Statement of the Problem:

Emergency department triage is the single highest-leverage decision point in emergency care. The triage clinician's initial acuity assignment determines the speed of intervention, the allocation of beds and diagnostic resources, and, in time-critical conditions such as sepsis, stroke, and traumatic brain injury, whether a patient survives. Conventional triage systems including the Emergency Severity Index (ESI) and Manchester Triage System (MTS) rely on subjective clinical assessment. Under high patient volumes, that assessment is degraded by fatigue, cognitive overload, and inter-rater variability.

Machine learning models have demonstrated consistent superiority over conventional triage systems in retrospective datasets from the United States, South Korea, Taiwan, and Portugal, with diagnostic accuracy indices (AUROC) ranging from 0.85 to 0.99 for critical-care outcome prediction (Tyler et al., 2024). Araouchi and Adda (2024) further demonstrated that stacking ensemble models can achieve 80% accuracy in triage-level assignment with an AUC of 0.93, using only structured variables collected at initial presentation.

However, the entirety of this evidence base is retrospective, drawn from high-resource settings with mature electronic health record infrastructure, and validated in demographically narrow populations. No published study has evaluated AI-assisted triage in a Caribbean public hospital. The applicability of existing models to a Jamaican ED context, its demographics, case mix, data infrastructure maturity, and staffing ratios is entirely unknown. Furthermore, documented cases of racial bias in healthcare AI models raise specific equity concerns for a predominantly Afro-Caribbean patient population (Da'Costa et al., 2025). This project addresses that gap directly.

## Previous Work:

Machine learning applications to emergency triage have been extensively studied in high-income healthcare systems over the past decade.

Tyler et al. (2024) conducted a systematic scoping review of 29 US-based primary studies (2013–2023), applying formal JBI critical appraisal before inclusion. ML models outperformed the ESI across all key outcome metrics: critical-care prediction, hospital admission forecasting, mistriage rate reduction, and mortality risk stratification. The most accurate single model achieved an AUROC of 0.991 for critical hospitalisation in a paediatric cohort. XGBoost and deep neural networks were the most frequently cited high-performing algorithms. Critically, all 29 included studies were retrospective, and the authors call explicitly for prospective RCTs and multi-site validation before clinical deployment.

Da'Costa et al. (2025) synthesised broader implementation evidence across 2014–2024 literature, identifying real-time analytics, natural language processing of clinical notes, and wearable device integration as key functional pillars of next-generation AI triage systems. Reported operational gains include up to 30% reduction in patient wait times and up to 20% reduction in time-to-treatment. The authors also surface the most substantive equity concern in the reviewed literature: at least one deployed AI healthcare system has been documented to systematically underestimate illness severity in Black patients, resulting in delayed care. They call for mandatory bias auditing, diverse training datasets, and human-in-the-loop governance as prerequisites for responsible deployment.

Araouchi and Adda (2024) provide direct empirical evidence through the TriageIntelli study, benchmarking seven ML algorithms against Korean Triage and Acuity Scale (KTAS) labels in 1,267 patients across two Korean EDs. Their stacking ensemble model (SVM + GBM + logistic regression) achieved 80.05% accuracy and an AUC of 0.93 — outperforming XGBoost (76.9%) and all individually trained models. The authors reframe the central research question: rather than predicting patient outcomes, accurate triage-level assignment at the point of initial assessment is the foundational problem, since downstream error propagates from that first classification. Key limitations include single-dataset reliance, narrow geographic scope, and the use of SMOTE oversampling which may inflate reported metrics.

Across all three papers, no study addresses a Caribbean or low-to-middle-income country context. This is both a gap and a justification for the proposed work.

### **Proposed Solution:**

This project proposes a two-phase feasibility study conducted at Mercer General Hospital Emergency Department.

#### **Phase 1 : Data Infrastructure Audit and Retrospective Modelling (Months 1–4)**

A structured audit of Mercer General's existing triage records will establish data completeness, variable availability (vitals, chief complaint, assigned triage level, disposition, outcome), and the minimum dataset size available for retrospective ML training. If sufficient structured data exists (target:  $\geq 5,000$  complete patient encounters), a baseline ML model will be trained on Mercer General's own data using the algorithm families identified as strongest in the literature: XGBoost, Random Forest, and stacking ensembles, and evaluated against existing triage assignments using cross-validation. This phase will also map current IT infrastructure against the requirements for real-time AI integration.

#### **Phase 2 : Bias Audit Framework Development (Months 5–8)**

Drawing on the equity concerns raised by Da'Costa et al. (2025), a bias audit protocol will be developed and piloted against the Phase 1 model output. Audit dimensions will include performance parity across age groups, sex, and presenting chief complaint category, and, where data permits, socioeconomic proxy variables. The deliverable will be a reusable bias audit checklist suitable for application to any AI triage tool considered for deployment at Mercer General, regardless of vendor or algorithm, alongside a feasibility report with a go/no-go recommendation for a prospective pilot study.

The project's primary output is not a deployed AI system. It is an evidence-based feasibility assessment and evaluation framework: the prerequisite that Tyler et al. (2024) make clear must precede any responsible clinical deployment.

- Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*, 16(5), e59906. <https://doi.org/10.7759/cureus.59906>
- Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, 197, 105838. <https://doi.org/10.1016/j.ijmedinf.2025.105838>
- Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. *Procedia Computer Science*, 251, 430–437. <https://doi.org/10.1016/j.procs.2024.11.130>